Held-out interpolation does not validate zero-noise extrapolation: a 513× amplification of pure shot noise on a noiseless model

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(Dated: October 20, 2026)

Zero-noise extrapolation fits a noise-scaling curve to measurements taken at amplified noise and evaluates it at zero. Implementations are commonly validated by holding out one amplified-noise point and checking the fit predicts it. We show this validation is systematically blind to the failure mode that matters, and quantify the gap. In a per-Pauli extrapolation over 756 curves, the held-out prediction error was 7.5×10^{-4} on quantities bounded to $[-1, 1]$ — an excellent fit by any conventional standard — while the same fits, evaluated at the extrapolation target, turned a 0.0652 kcal/mol energy error into 33.48 kcal/mol, an amplification of 513×. The input carried *no* device noise: it was multinomial shot noise drawn from exact statevector probabilities for a noiseless model, so the entire discrepancy was manufactured by the estimator. We trace the mechanism to the variance inflation factor of the fit evaluated off its data support. At the held-out point this factor is below unity for three of the four model classes used; at the extrapolation target it reaches 4.0, 6.9×10^{7} and 1.7×10^{20} for the quadratic, exponential and rational classes. Shot noise then drives model-class selection toward the flexible nonlinear families — chosen for 62% of curves — precisely because they interpolate best, and those are the classes whose extrapolation is worst conditioned. A layer-by-layer control shows circuits, measurement, reconstruction and the extrapolator's own algebra are each exact to 10^{-11} or better on noiseless input, isolating the pathology to the interaction of finite sampling with off-support evaluation. We recommend that any zero-noise extrapolation report its variance inflation at the evaluation point, and that validation be performed by extrapolating to a withheld *lower*-noise point rather than by interpolation.

I. INTRODUCTION

Zero-noise extrapolation (ZNE) is the most widely deployed error mitigation technique for near-term quantum processors [1–3]. Its logic is simple: deliberately amplify the noise in a circuit by a controlled factor λ, measure an observable at several $\lambda > 1$, fit a curve, and evaluate it at $\lambda = 0$.

The practical question is how one knows the fit is trustworthy. The natural answer, and the one we adopted, is cross-validation: withhold one of the measured amplification levels, fit to the rest, and check that the fit predicts the withheld point. If it does, the functional form is presumably adequate.

This paper reports that the check is close to worthless for its intended purpose, and explains why in a way we believe generalises. The withheld point lies *between* or at the edge of the remaining data; the evaluation point $\lambda = 0$ lies outside it, on the opposite side from every measurement. These are different statistical problems. A fit can be excellent at the former and catastrophically ill-conditioned at the latter, and the conventional validation cannot see the difference by construction.

We did not discover this by analysis. We discovered it because a production pipeline produced an energy error of 33.48 kcal/mol — some 500 times the uncorrected error — on an input that contained no hardware noise whatsoever, and we were forced to find out why.

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II. THE PIPELINE AND THE CONTROL EXPERIMENT

The observable is the ground-state energy of H_4 (linear chain, 1.0 Å, STO-3G, 4 electrons in 4 orbitals), computed through an entanglement-forged decomposition [4] into two 4-qubit registers. Noise is amplified by native-gate folding, $G \to G\,(G^{-1}G)^{k}$, applied to the two-qubit entangling gate, giving amplification levels $\lambda \in \{1, 3, 5, 7, 9\}$. The energy requires 756 Pauli expectation values across all preparation slots, and the implementation fits an independent noise-scaling curve to each.

Model selection is itself cross-validated, in two stages: a model class is chosen per curve from {linear, quadratic, exponential, rational} by held-out prediction error, then refit. This is more careful than a fixed functional form, and it is the specific reason the failure is as large as it is.

Before attributing the blowup to the extrapolator we eliminated every other component. Table I reports a layer-by-layer control on exact statevector input, where the correct answer is known in closed form at every amplification level.

Every layer is exact. Folding at $\lambda = 9$ reproduces $\lambda = 1$ to 10^{-14}; the reconstructed energy is correct to 5×10^{-11} kcal/mol at every level; and the two-stage extrapolator, fed exact data, selects the linear class with a held-out error of 4×10^{-16} and returns the exact answer. There is no bug in any of it.

TABLE I. Layer-by-layer control on noiseless statevector input. Each layer adds one stage of the production pipeline. Values are deviations from the exact result; folding is an identity operation on noiseless input, so the correct behaviour is independence from λ.

Layer	What it tests	Deviation
A	circuit construction, folding identity	$\leq 1.1 \times 10^{-14}$
B	Pauli expectation extraction	$\leq 1.3 \times 10^{-15}$
C1	per-slot state reconstruction	$\leq 2.0 \times 10^{-14}$
C2	energy assembly (kcal/mol)	$\leq 5.5 \times 10^{-11}$
D	the extrapolator's own algebra	6.1×10^{-12}

III. THE FAILURE

We then changed exactly one thing. Instead of exact probabilities we drew 100,000 multinomial samples per circuit from those same exact probabilities, over 8 seeds — statistically identical to a real submission to a noiseless simulator, and containing no device noise of any kind. The identical production code was run verbatim.

Table II is the whole paper in one place. The held-out error is 7.5×10^{-4} on quantities bounded to $[-1, 1]$: by the standard diagnostic, these are good fits. The same fits evaluated at $\lambda = 0$ produce an energy error 513 times the unmitigated one, and every bit of that 33.4 kcal/mol was created by the estimator, because the input had no device noise to remove.

Two secondary numbers show how violently the fits misbehave off support. 115 of 756 curves extrapolated outside $[-1, 1]$ and had to be rejected — a Pauli expectation value is bounded by ± 1 by construction — and the worst two returned 17,295 and −8,214. A fit that predicts a held-out point to four decimal places and then returns five orders of magnitude outside a hard physical bound is not mildly misspecified; it is being evaluated somewhere it carries no information.

IV. MECHANISM: VARIANCE INFLATION OFF THE DATA SUPPORT

The diagnosis is the conditioning of the fit at its evaluation point. For a linear-in-parameters fit with design matrix X over the measured amplification levels, the variance of the prediction at a point x is

$$\mathrm{Var}[\hat{y}(x)] = \sigma^2 \, x^\top (X^\top X)^{-1} x \equiv \sigma^2 \, \mathrm{VIF}(x), \qquad (1)$$

where σ^2 is the per-point measurement variance and $\mathrm{VIF}(x)$ is a variance inflation factor determined entirely by the design — the choice of amplification levels and functional form — and not at all by the data. It can be computed before any measurement is taken.

Table III evaluates it at the held-out validation point ($\lambda = 9$) and at the extrapolation target ($\lambda = 0$), for the four classes in use, with the nonlinear classes linearised about their fitted parameters.

The structure is unambiguous. At the held-out point the inflation factor is below unity for every class but one: the measurement is being *averaged*, and the validation looks good because interpolation genuinely is a well-posed problem on this design. At the evaluation point the same designs inflate the variance by factors reaching 10^{20}.

The two columns are not measuring the same thing, and no amount of agreement in the first constrains the second. This is the precise sense in which held-out interpolation fails to validate extrapolation: the diagnostic is evaluated where the problem is well-conditioned and the answer is taken where it is not.

A. Why shot noise makes it worse, not merely noisier

A subtler effect compounds this, and it explains the factor of 513 rather than a factor of a few. Model selection is driven by held-out *interpolation* error. The flexible three-parameter nonlinear classes interpolate best — they have more freedom to accommodate the particular realisation of shot noise in the fitted points — so selection prefers them: rational was chosen for 207 curves and exponential for 262, 469 of 756 in total (62%), against 143 linear and 144 quadratic. Those are exactly the two classes whose VIF(λ=0) is astronomical.

Finite sampling therefore does not simply add variance to the extrapolation. It actively steers model selection toward the classes whose extrapolation is worst conditioned, because the criterion used to select them rewards the behaviour that is orthogonal to the one that matters. The more shot noise, the stronger the preference. This is a selection pathology, not an estimation one, and increasing the shot count mitigates it only slowly.

B. The pathology is in the fitting functions, not the curve count

One natural objection is that fitting 756 independent curves is simply reckless, and that a parsimonious estimator would be safe. We tested this. H_4's forged state at Schmidt rank $K = 6$ lives in a six-dimensional Hamming-weight-two sector with real amplitudes and unit norm, leaving five free parameters, against 35 for a general density matrix on the same register. We built that five-parameter physical-manifold estimator and it is a substantial improvement at a *single* amplification level: 3.4× better on one device noise model and 11× better on the noiseless one, and it passes an adversarial control — fitted to shuffled or random data it returns errors of order 10^3 kcal/mol, so it is not covertly injecting the answer.

Extrapolating its five components across λ reproduced the same blowup. Reducing 756 curves to 5 did not help,

TABLE II. Pure shot noise through the production extrapolator: 756 curves, 100,000 shots, noiseless model. The input contains no device noise, so the correct $\lambda = 0$ answer equals the $\lambda = 1$ value and the entire extrapolated error is manufactured by the estimator. Held-out interpolation error is excellent on quantities bounded to $[-1, 1]$; the extrapolation evaluated from the same fits is not.

Held-out interpolation		Energy error		Off-support behaviour	
Stage-1 mean	8.19×10^{-4}	at $\lambda = 1$ (raw)	0.0652 kcal/mol	rejected, $\lvert\langle P\rangle\rvert > 1$	115/756 (15.2%)
Stage-2 mean	7.50×10^{-4}	at $\lambda = 0$ (extrapolated)	**33.48** kcal/mol	worst two extrapolations	17,295, −8,214
		amplification	**513**×	nonlinear class chosen	469/756 (62%)

TABLE III. Variance inflation at the validation point versus the evaluation point, for the design $\lambda \in \{1, 3, 5, 7, 9\}$. For the nonlinear classes the statistics are over the 28 representative curves fitted; means and medians are both given because the distributions are heavy-tailed.

Class	VIF(λ=9)	VIF(λ=0)	ratio
linear	0.600	0.825	1.4
quadratic	0.886	2.115	2.4
exponential (median)	0.577	4.008	6.9
exponential (mean)	0.516	6.94×10^{7}	1.3×10^{8}
rational (median)	0.460	903.5	2.0×10^{3}
rational (mean)	2385	1.71×10^{20}	7.2×10^{16}

because the problem was never the number of curves; it is that off-support evaluation of a fitted nonlinear function is ill-conditioned regardless of how few such functions there are.

V. WHAT DOES TRANSFER

The constructive counterpart is worth stating, because the conclusion is not that error mitigation is hopeless. Across this project every method that applies a *direct* correction — a forward model of the noise channel inverted at the measured amplification level, in the manner of probabilistic error cancellation [1, 5] — behaved stably, and every method that fits a curve and evaluates it off support failed. The distinguishing feature is not sophistication; it is whether the estimator is evaluated inside or outside the support of its data.

Direct corrections fail differently, and better. In a separate pre-registered experiment on a trapped-ion device, an analytic direct correction of this kind missed its target by a factor of 4.2 and was found, by refitting a single scalar blind to the reference energy, to under-correct by a factor of ≈ 1.8 [6]. That is a bounded, diagnosable, one-parameter discrepancy. It is a different species of failure from manufacturing 33 kcal/mol out of shot noise.

VI. RECOMMENDATIONS

1. **Report the variance inflation at the evaluation point.** VIF(λ=0) depends only on the amplification levels and the functional form, so it can and should be computed and published before any data is taken. A design with VIF(0) $\gg$ VIF(λ_{val}) is announcing in advance that its validation is uninformative.

2. **Validate by extrapolation, not interpolation.** Withhold the *lowest* amplification level and extrapolate to it from the higher ones. This tests the operation actually performed. It is strictly available whenever a conventional held-out check is, and it costs nothing extra.

3. **Enforce physical bounds as a hard gate, not a filter.** That 15% of curves extrapolated outside $[-1, 1]$, two of them by four orders of magnitude, was recorded as a rejection count rather than treated as a failure of the procedure. A procedure that violates a construction bound on one curve in seven has been falsified, and discarding the violations conceals that.

4. **Do not select model class by interpolation error** when the quantity of interest is an extrapolation. The criterion rewards the classes that extrapolate worst.

5. **Prefer direct correction where a forward noise model is available.** The amplification-and-extrapolation route trades a bounded modelling error for an unbounded conditioning one.

VII. LIMITATIONS

These results come from one observable, one molecule, one amplification design ($\lambda \in \{1, 3, 5, 7, 9\}$ by two-qubit gate folding) and one implementation. The VIF argument is general — it is a property of linear least squares — but the specific magnitudes in Table III belong to this design, and a different set of amplification levels would give different numbers. We make no claim that published ZNE results are wrong; many use fixed low-order functional forms, for which Table III shows the inflation to

be modest (1.4× linear, 2.4× quadratic). Our claim is narrower and, we think, unavoidable: the held-out interpolation check does not bound the extrapolation error, the gap can be twenty orders of magnitude, and shot-noise-driven model selection moves estimators toward the regime where it is largest.

VIII. CONCLUSION

A per-Pauli zero-noise extrapolation with excellent held-out interpolation error, 7.5×10^{-4}, amplified pure shot noise into a 33.48 kcal/mol energy error on a model containing no device noise at all, a factor of 513 above the unmitigated value. Every other component of the pipeline is exact to 10^{-11} or better. The mechanism is the variance inflation of a fitted curve evaluated outside the support of its data, which reaches 10^{20} for the flexible nonlinear classes that shot-noise-driven selection prefers precisely because they interpolate well.

Held-out interpolation and zero-noise extrapolation are different statistical operations, and success at the first is not evidence about the second. We recommend reporting VIF at the evaluation point and validating by extrapolation to a withheld lower-noise point, both of which are free.

ACKNOWLEDGMENTS

This work was supported by the IonQ Research Credits Program. The author thanks Vadim Karpusenko (IonQ) for technical guidance, and in particular for the native-gate diagnosis that made the folding control of Table I possible.

In accordance with arXiv's policy on generative AI tools, we disclose that a large language model (Anthropic Claude) was used as a coding and drafting assistant during this work: for software implementation, analysis scripting, and preparation of this manuscript. All scientific claims, experimental design decisions and interpretations are the author's own, and the author takes full responsibility for the content. All numerical results were produced by the committed code and are reproducible from the public repository.

DATA AND CODE AVAILABILITY

All code, result files and a complete iteration-by-iteration research log retaining every negative result are at https://github.com/Venkatallu11/quantum-chemistry-vqe. The control experiment of Table I is `vqe/task29a_ideal_control_diagnosis.py`; the shot-noise blowup of Table II is `vqe/task29a_layerD2_realistic.py`; the conditioning analysis of Table III is `vqe/task29g_extrapolation_conditioning.py`. Each writes the corresponding `*_results.json` from which the tables are taken directly.

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